

Chapter 73

Animating with Motion Paths

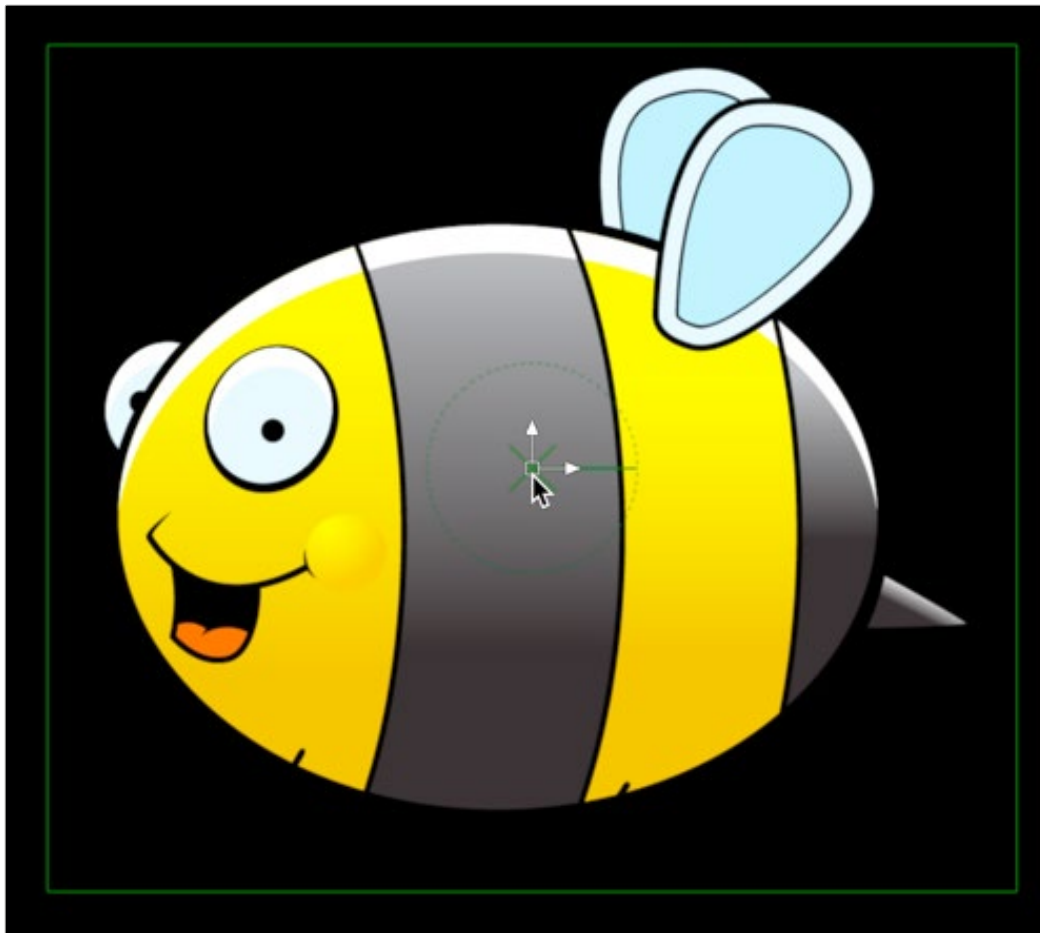
Layers and 3D objects can move along a designated spline shape to create motion path animations. This chapter discusses how you can create, edit, and use motion paths in Fusion.

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Animating Using Motion Paths

Motion paths are created from splines (polylines) including paint strokes and masks that have a path modifier applied to them. The path modifier defines the movement that transforms ordinary spline shapes into motion paths. You apply the path modifier either explicitly to the Center X/Y or Pivot X/Y parameters or implicitly by keyframing the Center or Pivot parameters. In either case, the spline path is displayed and then visually adjusted in the viewers. For all motion paths, the coordinate control in the viewer represents the position of an object or effect, such as a Transform or Rays node's center that moves along the path. Coordinate controls are represented onscreen with vertical and horizontal positioning arrows or an X.



A Center Offset onscreen control for an object's Transform node

The following nodes have parameters that can be animated using path modifiers to move an image around the composition. These include, but are not limited to:

- **Transform:** Center X/Y can be animated to move an image around.
- **DVE:** Center X/Y can be animated to move an image around.
- **Merge:** Center X/Y can be animated to move the Foreground connected image around.
- **Paint:** Stroke Controls > Center X/Y can be animated to move a stroke around.
- **Camera 3D node:** Translation X/Y/Z
- **Shape 3D node:** Translation X/Y/Z

The following nodes have parameters that can be animated using paths to alter the direction of a visual effect. These include, but are not limited to:

- **Directional Blur:** Center X/Y can be animated to change the direction of the blur.
- **Hot Spot:** Primary Center X/Y can be animated to move the hot spot around.
- **Rays:** Center X/Y can be animated to change the angle at which rays are emitted.
- **Polygon/BSpline/Ellipse/Rectangle/Triangle mask:** Center X/Y can be animated to move the mask.
- **Corner Positioner:** Top Left/Top Right/Bottom Left/Bottom Right X/Y can be animated to move each corner of the corner-pinned effect.
- **Vortex:** Center X/Y can be animated to move the center of the warping effect.

NOTE: It's not possible to add a motion path to a one-dimensional value, such as blur strength or merge angle. However, you can use the Spline Editor to edit these kinds of values in a visual way.

Types of Motion Paths

There are three types of motion paths: Polyline paths, XY paths, and 3D motion paths for 3D scenes.

- A Polyline path is generated by applying the path modifier. It uses two splines to define the path; one for the shape of the path displayed in the viewer, and a Displacement spline for the speed of the object along the path, displayed in the Spline Editor. The Polyline path is the default type of path modifier, and most documentation in this chapter assumes that this type is used.
- The XY path modifier employs a spline for the X position of the point and another for the Y position. The XY path modifier is explained in detail later in this chapter.
- 3D motion paths pertain only to positional controls within 3D scenes.

Polyline Path

Polyline paths are the easiest motion paths to work with. You can use the spline shape in the viewer to control the shape of the path, while a single Displacement curve in the Spline Editor is used to control the acceleration along the path. The most obvious way to create a Polyline motion path is by keyframing the Center X/Y parameter of a Transform node in the Inspector.

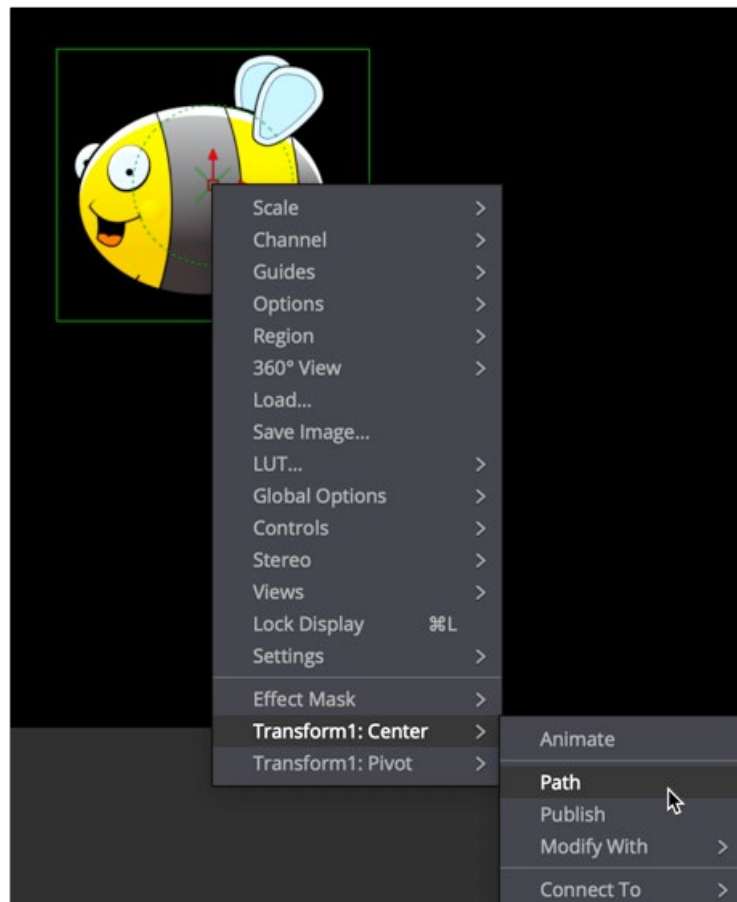
To create a Polyline motion path using the Center X/Y parameter in the Inspector:

- 1 Position the playhead on the frame where the motion will begin.
- 2 In the Inspector, click the gray Keyframe button to the right of the Center X and Y parameters.
This action applies the path modifier in the Modifiers tab in the Inspector.
- 3 Adjust the Center X and Y for the first keyframe position.
- 4 Position the playhead on the frame where the motion should change or stop.
- 5 In the Inspector, change the Center X and Y parameters to set a keyframe for the new location automatically.
- 6 In the viewer, modify and refine the motion path by selecting a control point and using any of the spline controls in the viewer toolbar.
- 7 Open the Spline Editor and adjust the Displacement spline to control the speed of the object along the path.

Keyframing Center X/Y is not the only way to apply the path modifier. An alternative method is to apply the path modifier to the Center X/Y parameter either in the Inspector or using the coordinate control in the viewer.

To apply a Polyline path to the Center X/Y parameter:

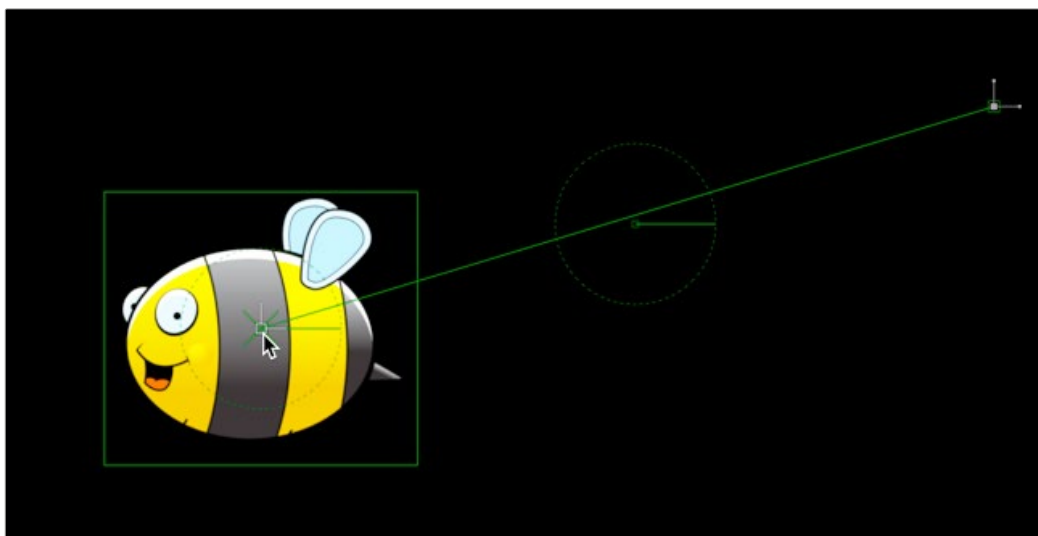
- 1 Position the playhead on the frame where the motion will begin.
- 2 Position the center coordinate control for the object, or effect, at its starting position.
- 3 Right-click the onscreen center coordinate control in the viewer (or over the Center X and Y parameter in the Inspector), and choose Path from the contextual menu for that control.



Applying a path modifier to the center of a Transform node

The object now has a path modifier applied, so without setting a keyframe you can drag the object to begin creating a motion path in the viewer.

- 4 Move the playhead to a new frame.
- 5 Drag the onscreen coordinate control or adjust the Offset or Center values in the Inspector. A keyframe is automatically created on the motion path, and a polyline is drawn from the original keyframe to the new one.



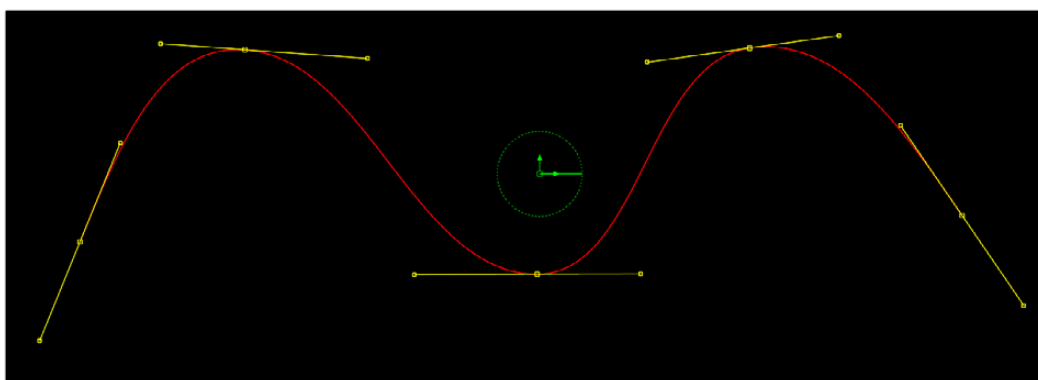
Two keyframes spaced several frames apart display a motion path showing the direction of animation.

- 6 The position of the center control is interpolated between the two keyframes. You can continue adding points by moving the playhead and adjusting the object's position until the entire motion path you need is created. For motion paths, there's no need to close the spline path; you can leave it open.
- 7 Upon completion, set the polyline to Insert and Modify mode by selecting a point on the path and pressing Command-I or clicking the Insert and Modify button on the toolbar. Don't worry too much about the overall shape of the motion path at this point. The shape can be refined further by adding additional points to the polyline and modifying the existing points.

A final alternative method for creating a motion path is to draw a spline shape first and then connect a path modifier to the spline. Using any of Fusion's spline tools, you can draw the shape of the path and then connect the path modifier to the published spline. Once the path modifier and the published spline are connected, you can keyframe the Displace parameter to move an image along the path. This method is useful when you want to use a paint stroke or mask shape as a motion path.

To use a mask shape or paint stroke as a path:

- 1 Draw a polygon mask or paint stroke in the shape of the path you want to use.
You do not have to close the mask shape.



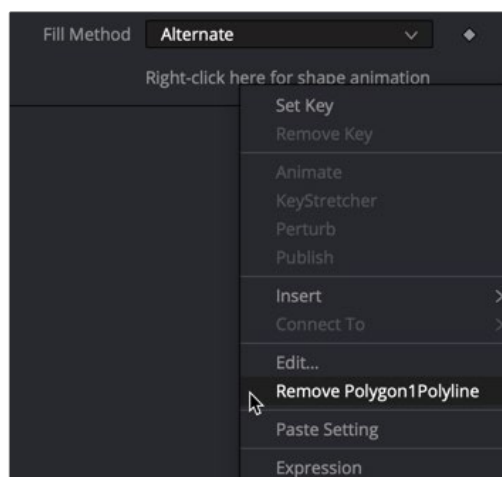
An open Polyline mask shape

- 2 When done drawing the shape, click the Insert and Modify button in the viewer toolbar to leave the mask shape as an open spline.

- 3 At this point you can select any of the control points along the spline and press Shift-S to make them smooth or Shift-L to make them linear.

All mask polylines have animation enabled by default, but that is usually not desirable for a motion path. You will need to remove this keyframe animation if you are using a mask shape.

- 4 At the bottom of the Inspector, right-click on the “Right-click here for shape animation” label and choose Remove Polygon1Polyline.



Right-clicking at the bottom of the Inspector to remove auto-keyframing from a mask shape

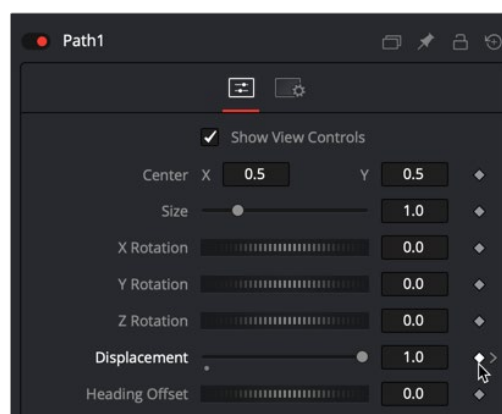
- 5 Right-click at the bottom of the Inspector again and select Publish to give other nodes access to this spline shape. (For a paint stroke, you will need to make the Stroke editable first by clicking the Make Editable button in the Stroke Controls.)

This enables the Modifiers tab with the Published Polyline modifier. This published spline can be used to define the shape of splines in other nodes.

- 6 Connect a Transform node to the image you want to have follow the path.
- 7 Right-click over the Center X/Y parameter in the Inspector and choose Path.

This adds a path modifier into the Modifiers tab.

- 8 In the Inspector, click the Modifiers tab and double-click the Path1 heading to open its parameters. The Displacement parameter already has a keyframe on it automatically. You'll want to remove that so you can set your own.
- 9 Click the red keyframe button to the right of the Displacement parameter to remove it.



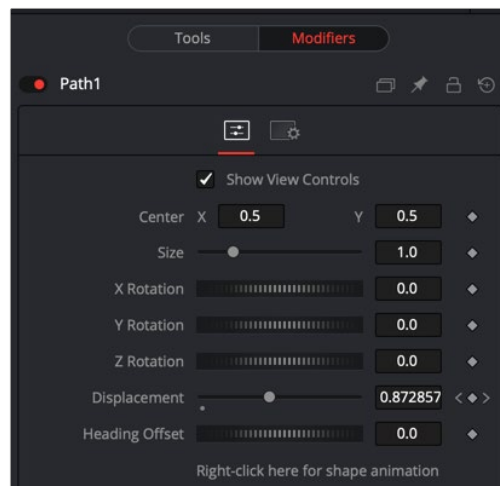
Remove the automatic displacement keyframe prior to creating your own keyframes.

- 10 At the bottom of the Modifiers tab, right-click on “Right-click here for shape animation” and choose Connect To > Polygon1Polyline.
- 11 To quickly see where your object has gone, drag the Displacement slider back and forth.
- 12 You may want to use the Size parameter to adjust the size of the overall path.

The Displacement slider is meant to be keyframed for animating the object along the path.

Path Modifier

In terms of functionality, it makes no difference which method you use to generate the path modifier. All the above methods are just different ways to get to the same point. Whichever way you decide to add the path modifier, the Modifiers tab contains controls for the path.



Creating a path adds controls to the Modifiers tab in the Inspector.

You can use the path modifier controls in the Inspector to change the position, size, and rotation of the entire path shape. The Displacement parameter is represented as a spline in the Spline Editor, which determines the object's position along the path and the Heading Offset is used for the orientation along the path.

Controlling Speed and Orientation along a Path

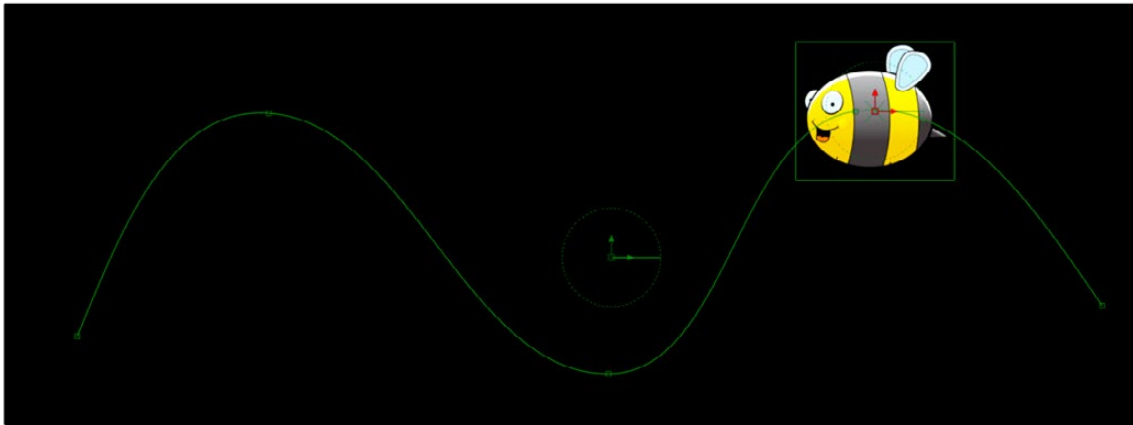
Every Polyline path has an associated Displacement spline in the Spline Editor. The Displacement spline represents acceleration, or the position of the animated object along its path, represented as a value between 0.0 and 1.0.



The Displacement curve of a Poly path represents the acceleration of an object on a path.

Smaller values are closer to the beginning of a path, while larger values are increasingly closer to the end of the path.

For instance, let's say you have a bumblebee that bobs up and down as it moves across the screen. To have the bee accelerate as it moves up and down but slow down as it reaches its peaks and valleys you use the Displacement curve.



A curvy path defined by a spline shape

The curved shape path does not define how fast the bee moves. The speed of the bee at any point along the path is a function of the Displacement parameter. You can control the Displacement parameter either in the Modifiers tab or in the Spline Editor.

To use the Displacement parameter to control the speed of an object on a path, do the following.

- 1 Position the playhead at the start of the animation.
- 2 In the Modifiers tab, drag the Displacement parameter to 0.0. This positions the object at the start of the path.
- 3 Click the Keyframe button to the right of the Displacement parameter.
- 4 Position the playhead somewhere further into the comp and drag the Displacement parameter until the object is where you want it to be based on the current frame.
- 5 Continue updating the playhead and the Displacement parameters at key points in the comp until you have reached the end of the path.

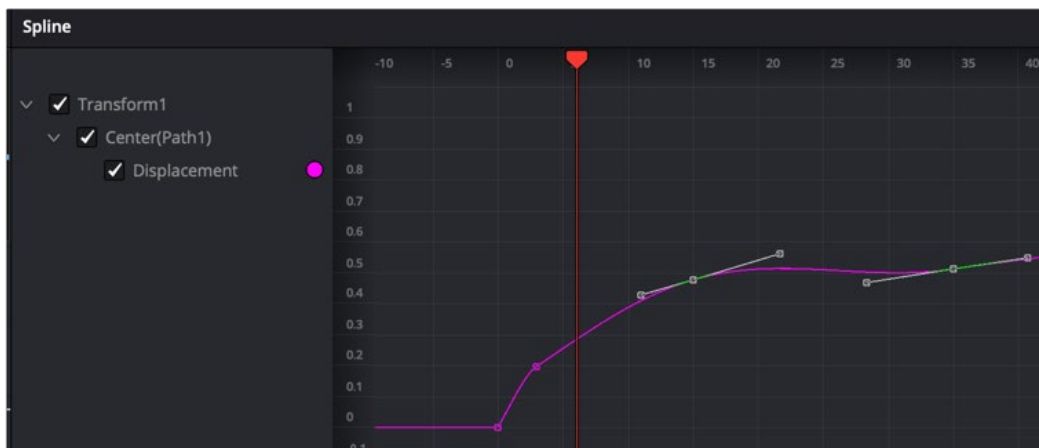
After the initial animation is set, you can use the Displacement curve in the Spline Editor to adjust the timing.

To adjust the timing of an object along a path:

- 1 Open the Spline Editor and enable the Displacement spline in the header.
- 2 Move control points horizontally closer together to increase the speed between the two points while maintaining the location of the object along the path.

A longer, flatter curve between two points indicates a slower rate of change.

- 3 Drag a control point up or down to change its location on the path while maintaining the timing between two points.



A Displacement curve in the Spline Editor

TIP: Holding down the Option key while clicking on the spline path in the viewer will add a new point to the spline path without adding a Displacement keyframe in the Spline Editor. This allows you to refine the shape of the path without changing the speed along the path.

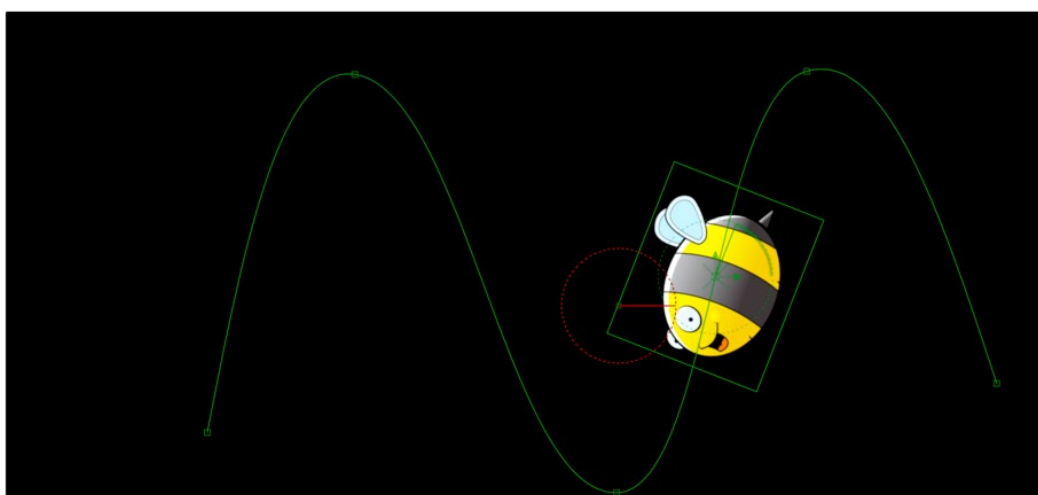
Using a Path Modifier to Adjust Orientation

The Heading parameter is used to adjust the orientation of the object along the path. For instance, if you want the bee in our animation to auto-orient based on the direction of the path, you can connect the bee's angle to the Heading parameter.

To connect the Heading to an object's angle:

- 1 Right-click over an object's angle parameter in a Transform node.
- 2 Choose Connect To > Path > Heading.

In our example comp, the bee now auto-oriens as the path descends and rises.



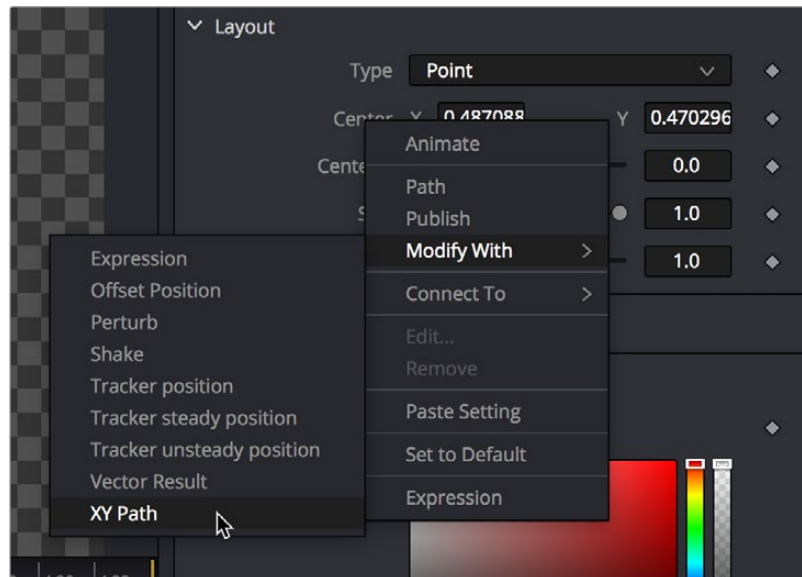
The Transform's angle parameter connected to the path modifier's Heading parameter

XY Path

Unlike a Polyline path, the XY path modifier uses separate splines in the Spline Editor to calculate position along the X-axis and along the Y-axis.

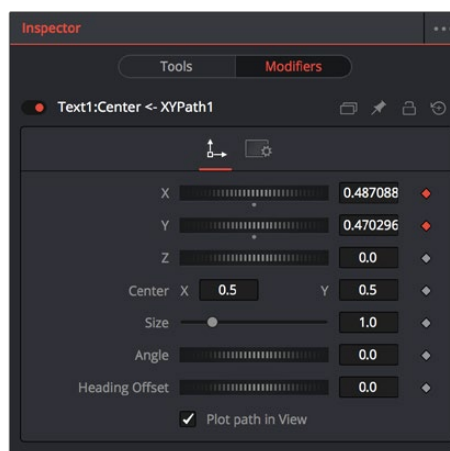
To animate a coordinate control using an XY path modifier:

- Right-click on the center coordinate control in the viewer or the Center X/ Y parameter in the Inspector, and then choose Modify With > XY Path from the contextual menu.



Adding the XY Path modifier to a Center parameter in the Inspector

At first glance, XY paths work like Polyline paths. To create the path once the modifier is applied, position the playhead and drag the onscreen control where you want it. Position the playhead again and move the onscreen control to its new position. The difference is that the control points are only there for spatial positioning. There is no Displacement parameter for controlling temporal positioning.



XY path modifier controls in the Modifiers tab of the Inspector

Instead of dragging in the viewer, you can use the controls in the Modifiers tab to create a motion path, while using the object's original Inspector controls as an offset to this motion path. You can use the XYZ parameters to position the object, the Center X/Y parameters to position the entire path, the Size and Angle to scale and rotate the path, and the Heading Offset control to adjust the orientation.



Using an XY path modifier to animate a piece of text

Using the XY Paths in the Spline Editor

The Spline Editor for the XY path displays the X and Y channel splines. Changes to the path in the viewer or the Inspector will be displayed as keyframes on these splines in the Spline Editor. Unlike a Polyline path, XY paths do not include a Displacement curve. The speed of the object along the path is tied to the path itself and cannot be separated from the timing of the keyframes that define that path.

TIP: XY path and Poly path can be converted between each other from the contextual menu. This gives you the ability to change methods to suit your current needs without having to redo animation.

The advantage of the XY path modifier is that you can explicitly set an XY coordinate at a specific time for more control.



XY Path displays X and Y curves in the Spline Editor but does not include a Displacement control.

Switching Default Paths

If you want to change the default path type to XY path, you can choose Fusion > Preferences > Globals in Fusion Studio or Fusion > Fusion Settings in DaVinci Resolve. In the Settings window, select the Defaults category and choose XY Path for the Point With drop-down menu. The next time you keyframe the Center X/Y parameter or choose Animate from CenterX/Y's contextual menu, an XY path modifier will be used instead of a Polyline path.

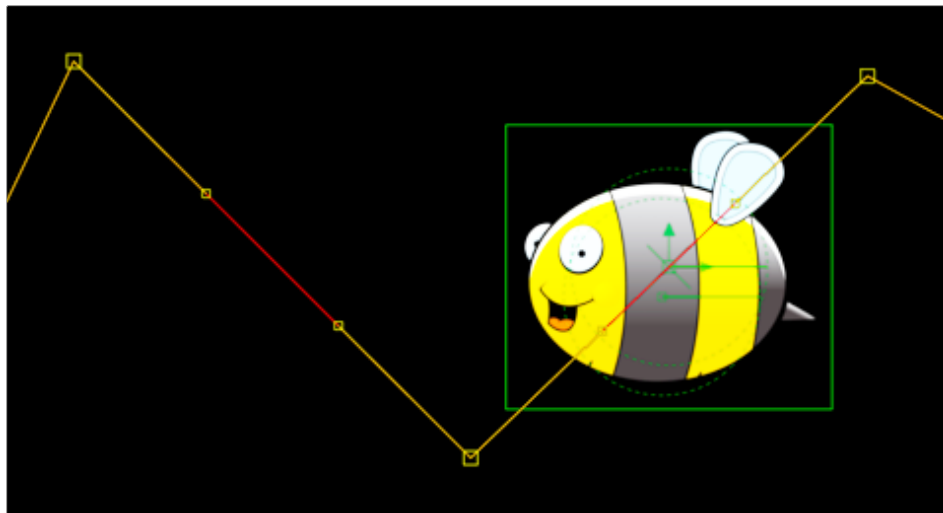
Types of Control Points

The control points along an XY path in the viewer are locked to the control points on the X and Y curve in the Spline Editor. The number of points are identical, and adding a control point in one place adds it to the other. That is not the case with a Polyline path. Polyline paths are composed of locked and unlocked points. Whether a point is locked is determined by how it was added to the Polyline. Locked points on the motion path in the viewer will have an associated point on the Displacement spline in the Spline Editor; unlocked points will not have corresponding points. Each has a distinct behavior, as described below.

Locked Points

Locked points are the motion path equivalents of keyframes. They are created by changing the playhead position and moving the animated control. These points indicate that the animated control must be in the specified position at the specified frame.

The locked points are displayed as larger-sized hollow squares in the viewer. Each locked key has an associated point on the path's Displacement curve in the Spline Editor.

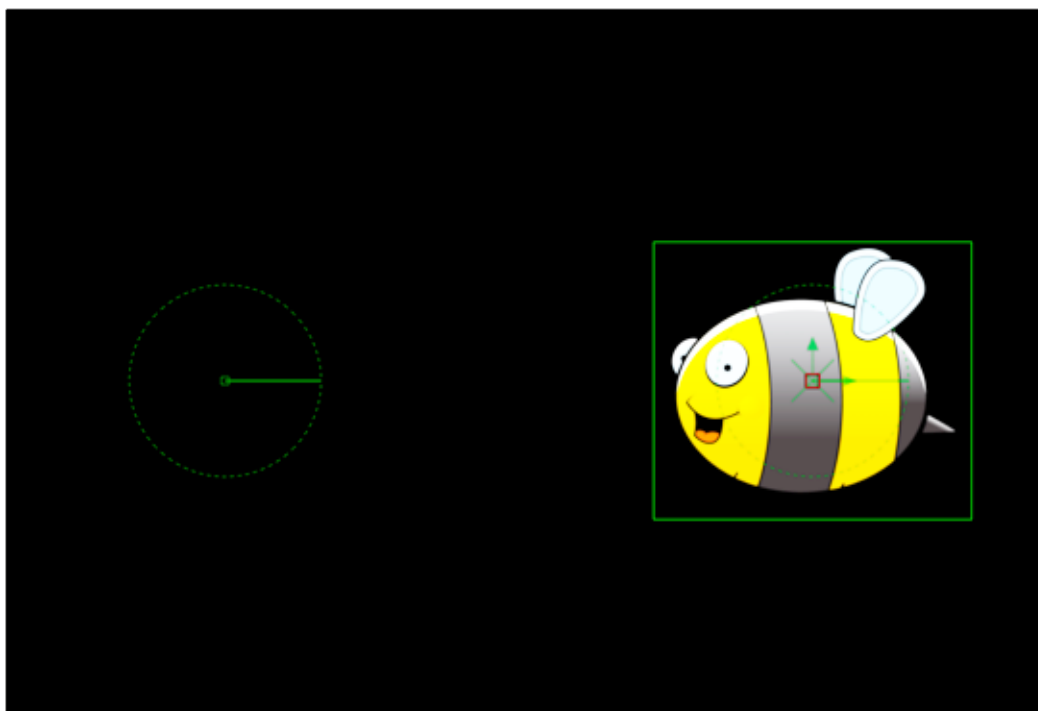


Locked points on a path in the viewer

Deleting a locked point from the motion path will change the overall timing of the motion.

To change the duration of a path using locked points:

- 1 Connect an object to a Transform node.
- 2 Position the where you want to start the motion path.

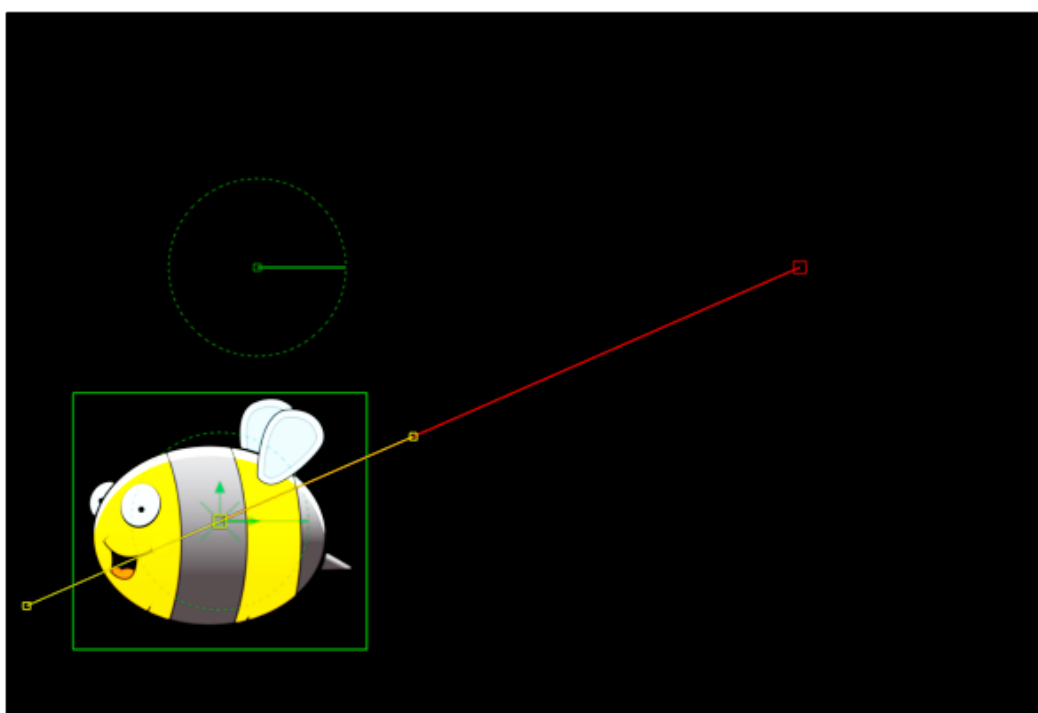


A graphic placed on the right side of the frame

- 3 Set the playhead at frame 0.
- 4 Select the Transform node and in the Inspector, click the Keyframe button to the right of the Center X/Y parameter.

This adds the path modifier and creates the first locked point of the path.

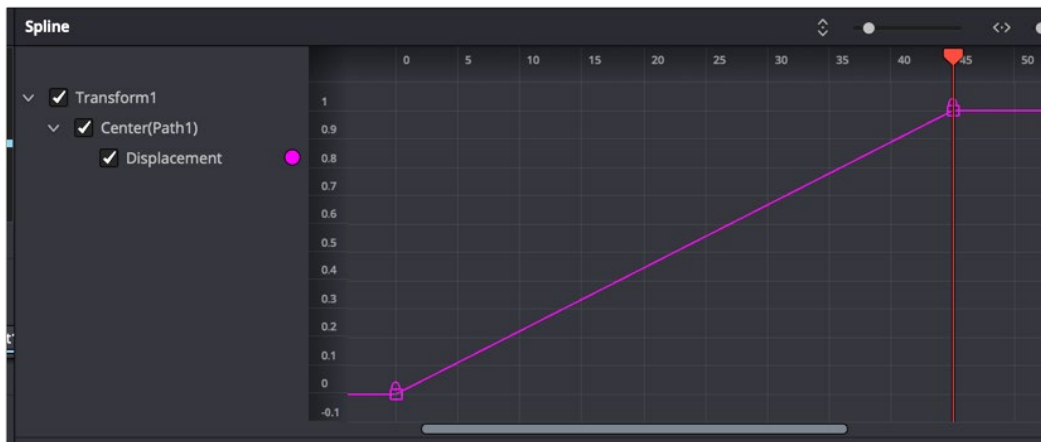
- 5 Position the playhead at frame 45.
- 6 Move the object's center to the lower center of the screen.



Moving the playhead and repositioning the bee adds a second locked point

This sets the second locked point.

- 7 View the Spline Editor and display Path1's Displacement spline.



The path's Displacement spline with locked keyframes

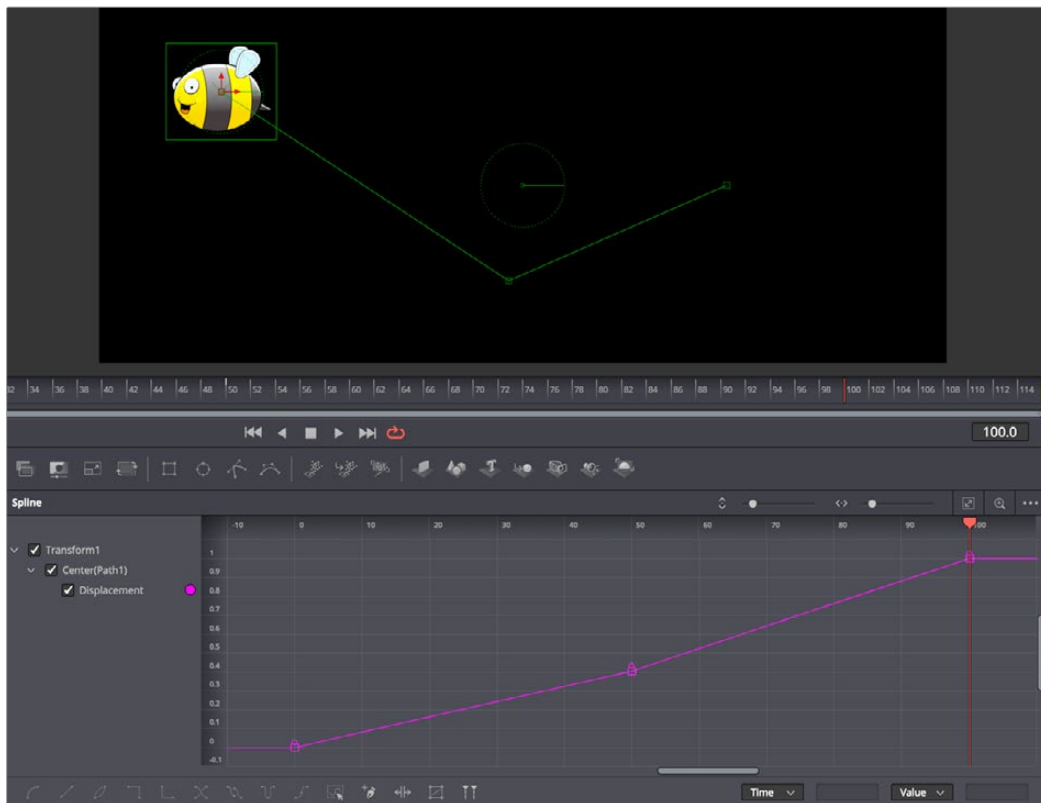
At a value of 0.0, the control will be located at the beginning of the path. When the value of the Displacement spline is 1.0, the control is located at the end of the path.

- 8 Select the keyframe at frame 45 in the Displacement spline and drag it to frame 50.

The motion path is now 50 frames long, without making any changes to the motion path's shape.

If you try to change this point's value from 1.0 to 0.75, it cannot be done because the point is the last in the animation, so the value must be 1.0 in the Displacement spline.

- 9 Position the playhead on frame 100 and move the bee center to the upper-left corner of the screen.



Moving locked points changes the duration of a motion path without changing its shape

This will create an additional locked point and set a new ending for the path.

Unlocked Points

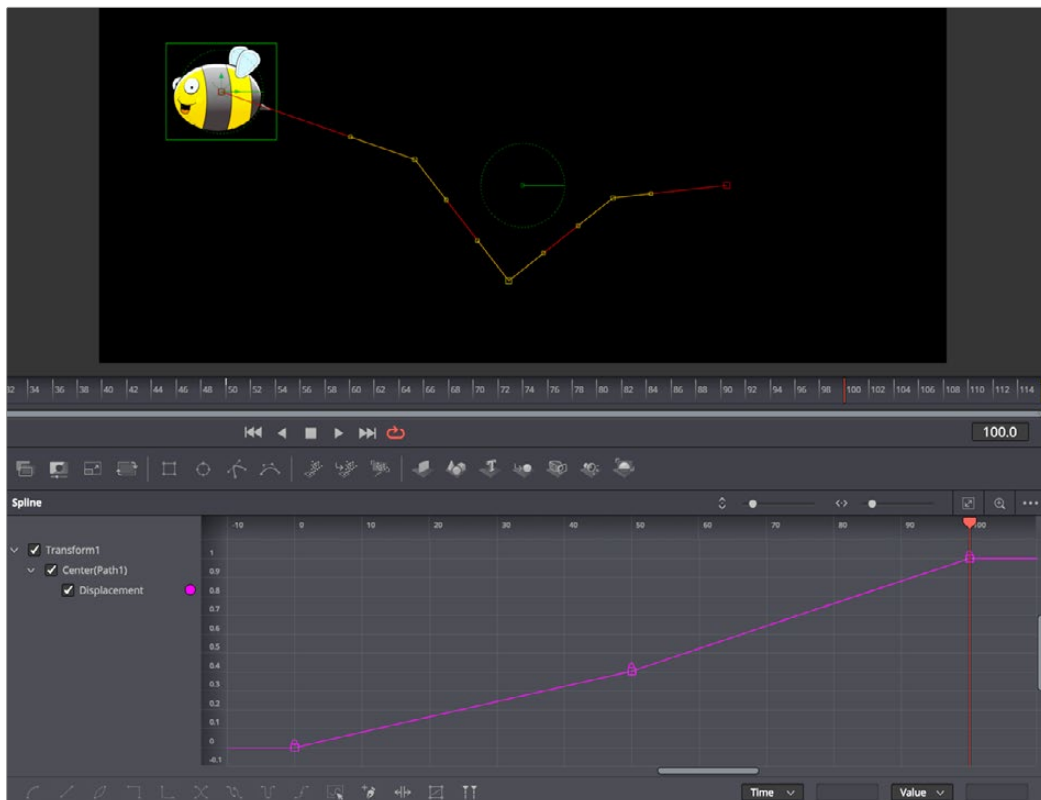
Unlocked points are created when additional points are added to the motion path while in Insert and Modify modes. These points are used to adjust the overall shape of the motion path, without directly affecting the timing of the motion. This means you can add whatever unlocked points you want to reshape a motion path, without that having any effect on the timing of the animation on that path. This makes it vastly easier to fine-tune a motion path spatially after you've perfected the timing of the animation temporally.

Unlocked points *do not* have corresponding points on the path's Displacement spline. They appear in the viewer as smaller, solid square points.

To add unlocked points to a motion path, do the following:

- 1 Select a motion path spline by using the Tab key to cycle controls until the path is selected.
- 2 To insert points along a path, click the Insert and Modify button in the toolbar.
- 3 Click on the path and create two new points: one half-way between the first and the second points, and the other half-way between the second and the third points.

The two points just added are not present in the motion path's Displacement spline. These are unlocked points, used to shape the motion but unrelated to the timing of the path.

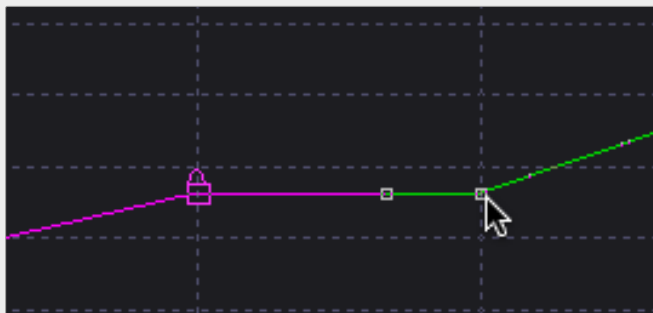


Unlocked points added to the motion path are not displayed on the Displacement spline

You can add unlocked points to the Displacement spline as well. Additional unlocked points in the Spline Editor can be used to make the object's motion pause briefly.

To pause motion along a motion path using an unlocked point:

Select a locked point on the Displacement spline and then hold down the Command key while dragging the point horizontally to another frame. The point is copied as an unlocked point to the new frame.



Create a pause in the motion by copying locked points

Knowing the difference between locked and unlocked points gives you independent control over the spatial and temporal aspects of motion paths.

Locking and Unlocking Points

You can change an unlocked point into a locked point, and vice versa, by selecting the point(s) and choosing the Lock Point option from the contextual menu.

Tips for Manipulating Motion Paths

There are a variety of ways you can create and edit motion paths in the viewer.

Compound Motion Paths Using Path Centers

Every motion path has a defined center represented by a crosshair. Path centers allow paths to be connected to other controls and behave in a hierarchical manner, which is an exceptionally powerful way of creating complex motion by combining relatively simple paths.

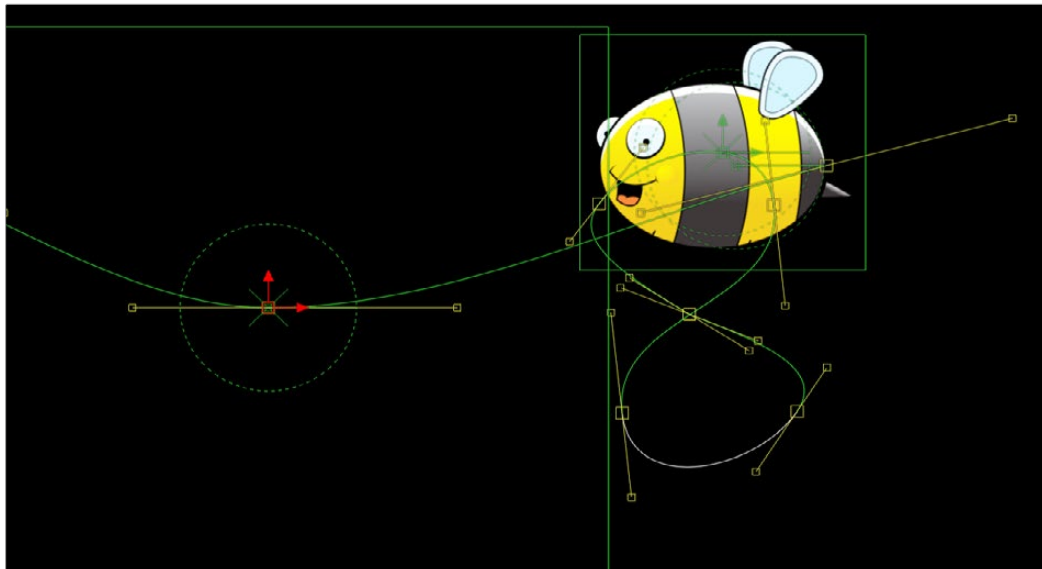
A useful example of this technique would be animating the path of a bee in flight. A bee often flies in a constant figure eight pattern while simultaneously moving forward. The easy way of making this happen involves two paths working together.

In the following example, the bee would be connected to a first path in a Transform node, which would be a figure eight of the bee moving in place. This first path's center would then be connected to another path defining the forward motion of the bee through the scene via a second Transform node.

To create a compound motion path:

- 1 Create a figure-eight motion path by keyframing an object or drawing a polyline mask.
(If using a polygon mask, you'll need to remove the auto-keyframing and publish the mask prior to the next step.)
- 2 Add a polyline mask and create a smooth curve spline that travels across the screen.

- 3 At the bottom of the Inspector, right-click over the “Right-click here for shape animation” label and choose Remove Polygon Polyline to remove the auto-animation behavior.
- 4 Right-click over the label again and choose Publish.
- 5 Select the object’s Transform node and click the Modifiers tab.
- 6 Right-click over the Path1 Center X/Y parameter and choose Path.
- 7 At the bottom of Path2, choose Connect To > Polygon: Polyline.
- 8 Keyframe the Path2 Displacement to move the object along the second path.



Two motion paths working together

Copying and Pasting Motion Paths

It is possible to copy an entire motion path to the clipboard and then paste it onto another node or path or composition.

Methods of copying and pasting motion paths:

- **To copy a motion path:** In the Inspector’s Modifiers tab, right-click on the path’s control header and choose Copy from the contextual menu.
- **To cut a motion path out of a node:** In the Inspector, right-click on the path’s control header and choose Cut from the contextual menu.
- **To paste the copied path over another path:** In the Inspector, right-click on the path’s control header and choose Paste from the contextual menu.

When pasting a path, the old motion path will be overwritten with the one from the clipboard.

Removing Motion Paths

There are multiple ways to remove or delete a path, and they all involve a right-click contextual menu. Removing a path modifier does not remove the object or the spline shape; it only removes the animation from the object and the modifier from the Modifiers tab in the Inspector.

To remove a path, do one of the following:

- In the Modifiers tab, right-click over the Path1 header and choose Delete from the menu.
- In the Inspector, right-click over the Center XY parameter and choose Remove Path1 from the menu.
- Right-click the center coordinate control in the viewer for the object you're animating, and choose Remove Path1 from the submenu of the NameOfObject; Center submenu.



Removing an entire motion path at once

Recording Motion Paths

You can animate both of the control's spatial and temporal information at the same time using the Record mode. This is useful when both position and speed are crucial to achieve the desired result.

Right-click on the desired path and select Record from the contextual menu. This displays a submenu of available data that may be recorded.

Use the Record Time option in conjunction with the Draw Append mode to create complex motion paths that will recreate the motion precisely as the path is drawn.

The time used to record the animation may not suit the needs of a project precisely. Adjust the path's Displacement spline in the Spline Editor to more correctly match the required motion.

Importing and Exporting Polylines

You can import and export polyline shapes into a common editable ASCII text file or its native format. These methods are used to save a particularly useful or generic mask or path for future use or for use in another application, such as Maya or LightWave. You can also import FXF, SSF, or Nuke shape files.

Native Format

To save a polyline shape in Fusion's native ASCII format, right-click on the header of the Mask node in the Inspector and select Settings > Save As from the contextual menu. Provide a name and path for the saved file and select OK to write a file with the .setting extension. This file will save the shape of a mask or path, as well as any animation splines applied to its points or controls.

To load the saved setting back into Fusion, you first create a new polyline of the same type, and then select Settings > Load from the mask's context menu or drag the .setting file directly into the Node Editor.

If you want to move a polyline from one composition to another, you can also copy the node to the clipboard, open your second composition, and paste it from the clipboard into the new composition.